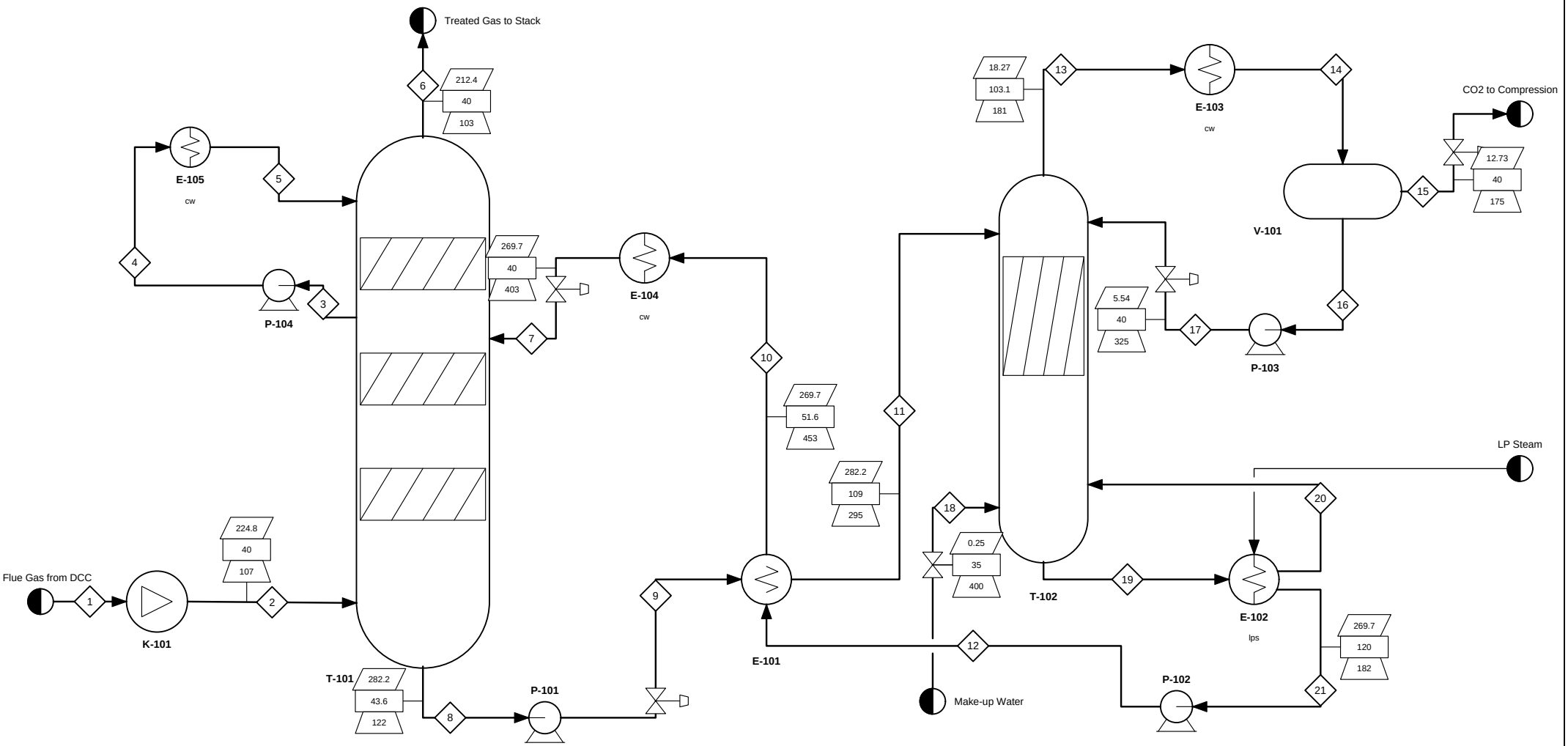
K-101 Flue Gas Blower	E-105 Wash Water Cooler	P-104 Wash Water Pump	T-101 Absorber + Water Wash	P-101 Rich Amine Pump	E-104 Lean Amine Cooler	E-101 Lean/Rich Exchanger	T-102 Stripper	E-103 Overhead Condenser	V-101 Reflux Drum	P-103 Reflux Pump	E-102 Kettle Reboiler	P-102 Lean Amine Pump



DESIGN POINT (MODEL)

Capture 90.0 % · CO2 product 12.7 t/h (wet) · SRD 3.94 GJ/t

E-101 18.4 MW · E-102 13.7 MW · E-103 4.1 MW · E-105 5.8 MW

K-101 313 kW · lps 22.94 t/h, 138.9 °C, 350 kPa a

NOTES

1. Rev 0 — issued for teaching. Flows t/h; temperatures °C; pressures kPa a.
2. Flag numbers come from the OTS simulator model at its design point (noise off). Training values, not a vendor design.
3. Flags only on streams the model calculates. Utilities: cw 30 °C, lps 3.5 bar a (assumed, TBC).
4. Tags as P&ID CC-PID-101. Only the process-defining control valves are shown. Sizes indicative.

LEGEND

Mass Flow (t/h) · Temperature (°C) · Pressure (kPa a) · Process Line · Utility Line

Control Valve · Process Input / Output · Stream Number · Line Crossing · Blower / Fan

Heat Exchanger · Centrifugal Pump · Packed Column · Vessel / Drum

MEA CO2 CAPTURE OTS VER 1 · CHEM ENG CHULALONGKORN UNIVERSITY

PROCESS FLOW DIAGRAM

MEA CO2 CAPTURE UNIT — ABSORBER T-101 / STRIPPER T-102

DRAWING No.	REV.	SIZE	SHEET
MEA-PFD-001	0	A4	1 OF 1
DRAWN	**CHECKED**	**SCALE**	**DATE**
Apinan Soottitantawat		NTS	2026-10-28